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Journal of Sport and Health Science 13 (2024) 793–804

Review

Risk factors for running-related injuries: An umbrella systematic review

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Received 12 September 2023; revised 17 January 2024; accepted 12 March 2024
Available online 30 April 2024

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Abstract

Purpose: This umbrella systematic review (SR) of SRs and meta-analysis seeks to comprehensively synthesize existing literature to identify and consolidate the diverse range of risk factors contributing to running-related injuries (RRIs).
Methods: Systematic searches were conducted on June 28, 2023, across Web of Science, SPORTDiscus, Scopus, PubMed, and Cochrane Library. We included SRs, whether accompanied by meta-analyses or not, that focused on investigating risk factors for RRIs within observational studies. The methodological quality of the SRs was evaluated using the Assessing the Methodological Quality of Systematic Reviews II. To assess the extent of overlap across reviews, the corrected covered area metric was calculated.
Results: From 1509 records retrieved, 13 SRs were included. The degree of overlap between SRs was low (4%), and quality varied from critically low ($n = 8$) to low ($n = 5$). Two hundred seven outcomes assessed in 148 primary studies were identified as being associated with the occurrence of RRIs. The effect sizes of the associations for which risk measures were reported ($n = 131$) were classified as large ($n = 30$, 23%), medium ($n = 38$, 29%), small ($n = 48$, 37%) or no effect ($n = 15$, 11%). Running/training characteristics, health and lifestyle factors, along with morphological and biomechanical aspects, exhibit large effect sizes in increasing the risk for RRIs.
Conclusion: Drawing from the outcomes of the low-quality SRs and associations with large effect sizes, our findings indicate that running/training characteristics and health and lifestyle factors, as well as morphological and biomechanical aspects, are all implicated in elevating the risk of RRIs, emphasizing the multifactorial basis of injury incidence in running. Given the low quality and heterogeneity of SR, individual findings warrant cautious interpretation.

Keywords: Umbrella review; Runners; Injury; Risk factor

1. Introduction

Running is one of the most popular and practiced sport activities.[1] It is an accessible, low-cost sport with well-known health benefits that has attracted many individuals that choose running as their main sport activity.[1,2] Running-related injuries (RRIs), however, are common among runners[3–5] with an incidence rate that varies from 4.2[6] (recreational runners) to 33.0[7] (novice runners) per 1000 h of running exposure. The knee, lower leg, and foot are the most affected sites,[3] whereas patellofemoral pain syndrome, medial tibial stress, plantar fasciitis, iliotibial band syndrome, and Achilles tendinopathy are the most frequent diagnoses.[3,4] RRIs lead to excessive health costs that often include visiting a general practitioner and medical specialist, surgical treatment, physiotherapy, and medication.[8,9] Time loss to recovery[10] and absences from work[8,9] also have an important impact on the individual quality of life and productivity.

RRIs are multifactorial and result from the complex interaction between individual attributes (e.g., age, gender, body mass index),[4,11,12] morphological risk factors (e.g., navicular drop, Q-angle, and arch index),[12–14] biomechanics (e.g., running kinematics and kinetics),[15–17] and running/training features (e.g., training frequency, volume, experience),[4,12] as well as health and lifestyle factors (e.g., medical history, alcohol consumption and participation in other sports).[4,11,14] These 5 domains have

Peer review under responsibility of Shanghai University of Sport.
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https://doi.org/10.1016/j.jshs.2024.04.011
Cite this article: Correia CK, Machado JM, Dominski FH, de Castro MP, de Brito Fontana H, Ruschel C. Risk factors for running-related injuries: An umbrella systematic review. *J Sport Health Sci* 2024;13:793–804.

received great attention but despite extensive research, risk factors for RRIs remain unclear.[14,18]

A comprehensive analysis integrating previously published systematic reviews (SRs) or meta-analyses is missing.[6,11,13,14,16–20] In the last 3 years, 7 SRs and meta-analyses were published, including 155 observational prospective studies.[11,13,17,19–22] To identify and consolidate the diverse range of risk factors contributing to RRIs, an umbrella review, and analysis of the current state of knowledge in this field is necessary. Current knowledge on this topic will support evidence-based practice and guide future studies.[23,24] In this study, we aimed to examine risk factors for RRIs through an umbrella SR of SRs and meta-analysis.

2. Methods

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines[25] and recommendations for Umbrella Reviews[23] were followed to conduct this study. The protocol of this study was registered in the International Prospective Register of Systematic Reviews (PROSPERO) (registration number CRD42021286241).

2.1. Search strategy

Five electronic bibliographic databases were searched: Web of Science, SPORTDiscus, Scopus, PubMed, and Cochrane Library. We limited searches to peer-reviewed journal articles published in English language from database inception to June 28, 2023. The search strategy used keywords related to injury, running, and reviews. Full description of search strategy used in each database is presented in Supplementary File 1.

The search strategy was implemented in the fields of title, abstract, and keywords of each database. The articles resulting from the search were inserted into Endnote (Version 21; Clarivate, Philadelphia, PA, USA), and duplicates were removed. Articles were then exported to Rayyan platform[26] and 2 independent researchers (CKC and JMM) conducted the study selection. A third researcher (CR) resolved any discrepancies.

2.2. Eligibility criteria

We used PECOS (Population, Exposure, Comparator, Outcomes, and Study design) strategy for eligibility criteria (Table 1).

2.3. Data extraction

Two authors (CKC and JMM) extracted data from 13 SRs. The results were similar and therefore, only 1 author (CKC) continued with data extraction. The following information was extracted from each SRs:

a) Publication year;
b) Geographic region: Country of first author's affiliation;
c) Review type: SR or SR with a meta-analysis;
d) Number of studies: Number of articles included under each review;
e) Publication period: Search period;
f) Search strategy: Databases searched;
g) Primary study type: Prospective, retrospective, or case-control;
h) Participants characteristics: Number of participants, age, gender, body mass index, running experience (novice, recreational, competitive, cross-country, short or long distance);
i) Follow-up period: Duration of participants' follow-up;
j) Injury definition: Whether the review reported the definition of injury of each primary;
k) Main anatomical injury site: Lower extremity, groin, knee, ankle, and/or other sites; [27]
l) Main diagnosis/injury: Patellofemoral pain syndrome, medial tibial stress syndrome, plantar fasciitis, Achilles tendinopathy, stress fracture, ankle sprain, and/or other diagnosis;[3]
m) Risk factors: Any outcome that was significantly associated ($p < 0.05$) with RRIs in a bivariate analysis; or that presented odds ratio (OR), relative risk (RR), or hazard ratio (HR) values greater than 1.0 in a regression analysis;
n) Measures of risk: OR, RR, or HR values; and 95% confidence interval (95%CI);
o) PRISMA: Whether the review followed the PRISMA statement;
p) Risk of bias: Methods used to assess risk of bias and the results of this analysis;
q) Level of evidence: Whether the review reported the level of evidence for each risk factor, which tool was used, and the interpretation of the result;
r) Meta-analysis: Estimates, effect size, and heterogeneity (I^2, as %).

Table 1
PECOS strategy.

PECOS strategy	Inclusion criteria	Exclusion criteria
Population	Healthy adults ≥18 years old	–
Exposure	Running	Other sports that involve running (i.e., soccer, handball, basketball, jogging, walking, *etc.*)
Comparator	No comparisons were made.	–
Outcomes	Risk factors for lower limb and/or lower back and/or trunk injury	–
Study design	Systematic reviews with or without meta-analysis that included observational studies (prospective, retrospective and case-control)	Case reports, comments, narrative reviews and not published in peer-reviewed journals

Abbreviation: PECOS = Population, Exposure, Comparator, Outcomes, and Study design.

Based on the extracted values of OR, RR, and HR, effect sizes of the measures were interpreted as follow[28]: no effect: RR or HR < 1.21 or OR < 1.31; small effect: RR or HR: 1.22−1.85 or OR: 1.32−2.37; medium effect: RR or HR: 1.86−2.99 or OR: 2.38−4.69; large effect: RR or HR > 3.00 or OR > 4.70.

A narrative synthesis of the included SRs was carried out according to the domains previously established in the literature[4,11,12]: (a) individual attributes, (b) running/training, (c) health and lifestyle, (d) morphological, and (e) biomechanical factors. For each domain, risk factors were identified based on the outcomes reported in the SRs.

2.4. Overlap

In umbrella reviews, overlap indicates the degree of primary studies were included in more than 1 SRs.[29,30] To identify the overlap 1 author (CKC) elaborated a citation matrix,[30] which is schematically shown in Fig. 1. The corrected covered area (CCA) was calculated using the following equation:[30]

$$CCA = \frac{N - r}{(r \times c) - r} \quad (1)$$

where N is the total number of included publications (total number of ticked boxes in the citation matrix); r is the number of primary studies (rows); and c is the number of SRs (columns).

The CCA results were expressed in percentage (%) and interpreted as follows: 0−5 as slight/low; 6−10 as moderate; 11−15 as high; and >15 as very high overlap.[30] A CCA of 100% means that every review included in this umbrella SR comprised the same studies. On the other hand, a CCA of 0% indicates that every review included was composed by unique studies.

Columns

Rows	Review 01	Review 02	Review 03
Primary study 01	X		X
Primary study 02	X	X	
Primary study 03	X	X	
Primary study 04	X		X

$$CCA = \frac{N - r}{(r \times c) - r} \qquad N = 8,\ r = 4,\ c = 3$$

$$CCA = \frac{8 - 4}{(4 \times 3) - 4} = 0.5$$

Fig 1. Example of matrix and corrected covered area (CCA) formula. c = number of systematic reviews (columns); N = Total number of included publications (total number of ticked boxes in the citation matrix); r = number of primary studies (rows).

2.5. Quality assessment

The quality assessment of the included SRs was performed using the Assessing the Methodological Quality of Systematic Reviews II (AMSTAR 2).[31,32] The checklist consists of 16 different items with 3 answering options: yes, partial yes, or no. Compared to the original checklist, the AMSTAR 2 was modified to cover studies of risk factors.[33] The items that refer to interventions were replaced by exposures.[33] The final rating was interpreted as follows: (a) high: no or 1 non-critical weakness; (b) moderate: more than 1 non-critical weakness; (c) low: 1 critical flaw with or without non-critical weaknesses; or (d) critically low: more than 1 critical flaw with or without non-critical weaknesses.[31,32] Two researchers (CKC and JMM) performed the quality assessment independently. Discrepancies were solved by discussion and consensus or by a third researcher (CR).

3. Results

The search identified 1509 records. After removing 731 duplicates and screening titles and abstracts, 83 SRs were considered for inclusion. Of the 83 studies selected for full reading, 43 studies had discrepancies. After the authors discussed the inclusion and exclusion criteria, the conflict between the 43 studies was resolved. Finally, 13 SRs were included after full-text assessment (Fig. 2). The reasons for exclusions are reported in Supplementary File 2.

The first SR was published in 2007.[4] The primary studies were mostly prospective, and the number of primary studies ranged from 1 study[19] to 66 studies.[21] Most SRs used Downs and Black Quality Index[17,19–21,34] to assess the risk of bias. Further methodological information from each SR is presented in Table 2.

Participants' characteristics, follow-up periods, and injury information are presented in Table 3. The number of included participants in each SR ranged from 52 participants[19] to 16,836 participants.[11] Seven SRs reported participant age, ranging from 18[35] to 50 years.[35] Participants' running experiences varied across categories such as novice,[12–14,25,36] recreational,[4,12–14,17,20,22,36] competitive[4,12–14,20] or experienced;[22,36] engaging in activities including cross-country,[12,14,17,19,20] track and field,[4] and short or long distances.[11] The follow-up periods, as reported in 7 SRs, ranged from 1 day[4] to 7 years.[20] The most prevalent diagnoses included overall RRIs,[11,12,14,17,20,36] Achilles tendinopathy,[11–13,17,21,36] patellofemoral pain,[11–13,17,21,36] iliotibial band syndrome,[13,17,21,35,36] and stress fractures.[11,19,36]

The 13 SRs included a total of 148 different primary studies that investigated risk factors for RRIs. The degree of overlap between SRs was considered low (CCA = 0.04; 4%). Findings are grouped and presented in detail in Supplementary Files according to the following domains: individual attributes (Supplementary File 3), running/training factors (Supplementary File 4), health and lifestyle factors (Supplementary File 5), morphological factors (Supplementary File 6), and biomechanical factors (Supplementary Files 7 and 8).

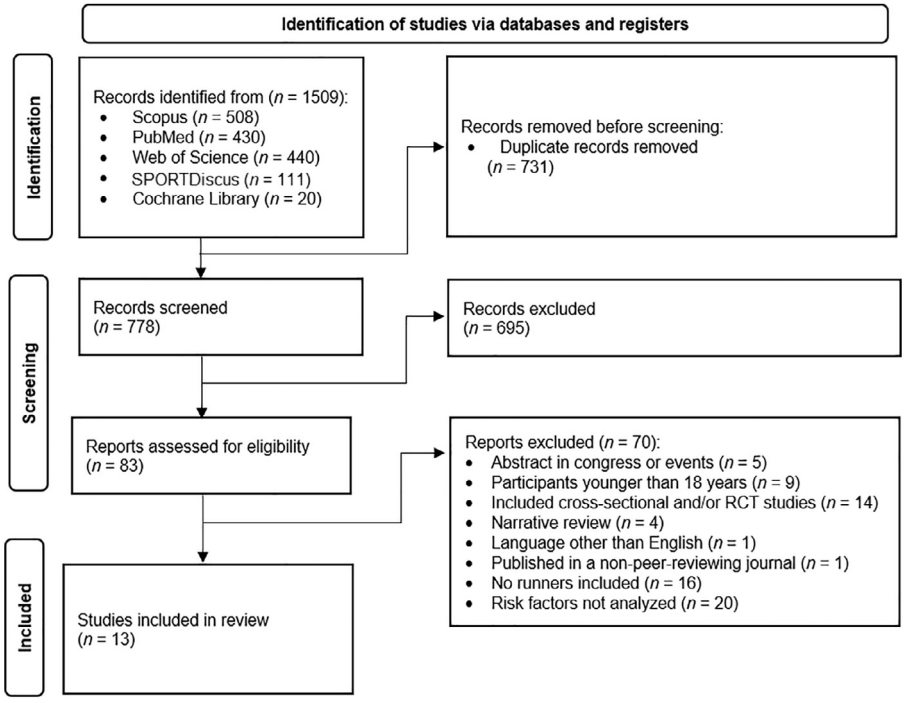


Fig 2. PRISMA 2020 flow diagram. PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses. RCT = randomized controlled trial.

The quality appraisal of the 13 SRs is presented in Table 4. In general, the quality of the SRs was classified as being critically low (8 reviews)[4,14,17,19,21,34–36] and low (5 reviews).[11–13,20,21]

Fifty-nine different factors across all domains were identified based on the outcomes reported in the SRs. Of these, 5 factors are from the individual attribute domain, 6 factors from the health and lifestyle domain, 17 factors from the running/training domain, 5 factors from the morphological domain and 26 factors from the biomechanical domain. These factors are listed in the first column of the respective Supplementary Files 3–8. The effect size of each outcome was reported, being classified as large (23% of the outcomes, $n = 30$), medium (29%, $n = 38$), small (37%, $n = 48$), and as no effect (11%, $n = 15$). For the remaining 38% ($n = 81$) of the outcomes reported in the SR, the value of the risk measure was not reported.

In Fig. 3, we summarize the risk factors for RRI. In this summary, we exclude outcomes associated with a RR or HR < 1.21 or OR < 1.31 (indicative of no effect) and also findings reported from critically low-quality SRs.

Four SRs performed meta-analysis.[13,17,19,34] Only biomechanical risk factors were investigated. During the stance phase of running, slower hip adduction velocity (mean difference = −12,8 °/s; $p = 0.04$),[13] higher peak force under 2nd metatarsal (standardized mean difference = 0.65; $p = 0.02$),[34] and under 3rd metatarsal (standardized mean difference = 0.60; $p = 0.03$);[34] and lower time to peak force underneath the lateral heel (standardized mean difference = −0.56; $p = 0.04$)[34] are shown to increase the risk for RRIs. Less knee extension strength (standardized mean difference = −0.19; $p = 0.03$)[13] was also considered to increase the risk for RRIs. Regarding the level of evidence available in each SRs, the results of the meta-analyses indicated limited and conflicting evidence for all these risk factors (Supplementary File 9).

4. Discussion

This umbrella SR of SRs and meta-analysis synthesizes the existing literature regarding the diverse range of risk factors contributing to RRIs, providing a comprehensive overview and analysis of the current state of knowledge in this field. This is the first umbrella review to systematically summarize risk factors for RRIs. Our results highlight the significant interest of researchers in this topic and corroborate the assumption of a multivariate etiology for RRIs.

Our analysis showed unsatisfactory quality for all the included SRs, ranging between critically low and low quality. This indicates the presence of methodological issues in their design, conduction and/or reporting, based on the critical domains assessed by AMSTAR 2 tool. Most SRs did not provide a list of excluded studies, failed to present a comprehensive literature search strategy, and did not provide a proper description of an explicit statement that the review methods were established prior (PROSPERO database registration and significant deviations from the protocol were stated). The registration of SRs minimizes the occurrence of duplicate reviews or reviews with similar objectives, promotes transparency, and reduces the potential risk of bias.[37] In agreement with tool recommendations,[31] reviews that were classified as critically low quality should not be used, and low quality SRs should be used with caution.[31] More high-quality SRs are necessary and will strengthen the evidence about RRIs to guide training and injury prevention programs.

Table 2
Methodological information from each SR.

Study	Geographic region	Review type	PRISMA	Databases searched	Search period	Primary studies	Number of primary studies	Risk of bias assessment tools	Level of evidence assessment tools
Van Gent et al (2007)[4]	Netherlands	SR	NR	PubMed	Databases inception until January 2006	Prospective and retrospective	17	Quality scoring tool elaborated by authors	Guidelines of van Tulder et al. (2003)[40]
Van Poppel et al. (2021)[11]	Netherlands	SR	Yes	PubMed, CINAHL, Cochrane Library	Databases inception until February 2019	Prospective	29	Quality in prognosis studies and Prediction model risk of Bias Assessment Tool	GRADE
Van der Worp et al. (2015)[12]	Netherlands	SR	No	CINAHL, Embase, PubMed and PsycInfo	CINAHL (1982 to December 26, 2012), Embase (1947 to January 1, 2013), PubMed (1940 to December 26, 2012), and PsycInfo (1806 to January 1, 2013)	Prospective and retrospective	15	Modified assessment tools of the Cochrane Collaboration and previous systematic reviews of risk factors for musculoskeletal disorders	Guidelines of van Tulder et al. (2003)[40]
Peterson et al. (2022)[13]	Australia	SR and meta-analysis	Yes	MEDLINE, Embase, CINAHL, SPORTDiscus, AMED and Cochrane library	Database inception until January 13, 2021	Prospective	30	Health Evidence Bulletins	NA
Saragiotto et al. (2014)[14]	Brazil	SR	NR	Embase, PubMed, CINAHL, SPORTDiscus, Latin American and Caribbean Centre on Health Sciences Information and Scientific Electronic Library Online	Embase (1980 to December 2012), PubMed (1946 to December 2012), CINAHL (1988 to December 2012) SPORTDiscus (1977 to December 2012), Latin American and Caribbean Centre on Health Sciences Information (1985 to December 2012) and Scientific Electronic Library Online (1998 to December 2012)	Prospective	11	Modified Newcastle Ottawa Scale	NA
Vannatta et al. (2020)[17]	USA	SR and meta-analysis	NR	MEDLINE, CINAHAL and PubMed	Databases inception to January 2019	Prospective	13	Downs and Black Quality Index	Modified version of guidelines of van Tulder et al. (2003)[40]
Anderson et al. (2020)[19]	Australia	SR	Yes	Medline, Embase, CINAHL and SPORT Discus	Databases inception to the 4th week of April 2019	Prospective[a]	53 (1)	Downs and Black Quality Index	Modified version of guidelines of van Tulder et al. (2003)[40]

(*continued on next page*)

Table 2 (Continued)

Study	Geographic region	Review type	PRISMA	Databases searched	Search period	Primary studies	Number of primary studies	Risk of bias assessment tools	Level of evidence assessment tools
Burke et al. (2021)[20]	Ireland	SR	Yes	MEDLINE, PubMed, SPORTDiscus and Web of Science	January 1960 to Nov 2020	Retrospective and prospective	13	Downs and Black Quality Index	GRADE
Willwacher et al. (2022)[21]	Germany	SR	Yes	PubMed	Databases inception to February 5, 2021	Prospective and retrospective	66	Downs and Black Quality Index	Guidelines of van Tulder et al. (2003)[40]
Ross et al. (2023)[22]	USA	SR		PubMed, Embase, CINAHL, Ovid MEDLINE	Database inception until November 30, 2020	Prospective and retrospective	10 (3)	Joanna-Briggs Institute Critical Appraisal Tool	NA
Neal et al. (2016)[34]	United Kingdom	SR and meta-analysis	Yes	Web of Science, MEDLINE and CINAHL	Database inception until April 2015	Case control and prospective	28 (2)	Downs and Black Quality Index	Guidelines of van Tulder et al. (2003)[40]
Louw and Dreary (2014)[35]	United Kingdom	SR	Yes	34 databases (main: CINAHL, PsycInfo, Embase, PubMed, ScienceDirect and Web of Science)	Databases inception to July 2011	Prospective and retrospective case control	12	Quality assessment tool for quantitative studies	Guidelines of Reid and Rivett (2005)[41]
Mann et al. (2016)[36]	Luxembourg	SR	Yes	PubMed, Embase, CINAHL, Science-Direct and Scopus	Database inception until March 2015	Prospective and retrospective	8	Modified version of assessment tool prepared by other authors	NA

Note: Numbers in brackets represent the number of only observational studies included in systematic review.
Abbreviations: GRADE = Grading of Recommendations Assessment, Development, and Evaluation; NA = not applied; NR = not reported; PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses; SR = systematic review.

Table 3
Participant's characteristics, follow-up period and injury information.

Review	Participants characteristics						Follow-up period (min−max)	Injury		
	Total participants	Average age (min−max)	Gender	Mean BMI (min−max)	Running experience	Other characteristics		Definition	Anatomical area	Main diagnosis
Van Gent et al (2007)[4]	13,662	NR	M/F	NR	Track and field, recreational and competitive	–	1 day−18 months	Yes	Ankle, lower leg, knee, upper leg, hip/pelvis and lower extremity	Skin lesions, pain/stiffness, overuse, cramps, ankle sprain, joint problems, tendonitis and other injuries
Van Poppel et al. (2021)[11]	16,836	36.6−43.3	M/F	24.6−26.3	Short and long distance	–	6 weeks−1 year	Yes	NR	Overall RRIs, PFP, Achilles tendinopathy, stress fractures
Van der Worp et al. (2015)[12]	9643	36−44	M/F	23.0−25.9	Novice, recreational, competitive and cross-country	–	8 weeks−1 year	Yes	NA	Overall RRIs, Achilles tendinopathy, and PFP
Peterson et al. (2022)[13]	3404	NR	M/F	NR	Novice, recreational and competitive	–	8 weeks−2 years	NR	NA	Lower leg pain, PFP, iliotibial band syndrome and Achilles tendinopathy
Saragiotto et al. (2014)[14]	4671	NR	NR	NR	Novice, recreational, competitive and cross-country	–	NR	NR	NR	Overall RRIs
Vannatta et al. (2020)[17]	932	19.2−45.0	M/F	19.9−24.5	Recreational and cross-country	–	1 season−2 years	Yes	NA	Overall RRIs, PFP, iliotibial band syndrome, Achilles tendinopathy
Anderson et al. (2020)[19]	52	20	M/F	20−21	Cross-country	Foot strike patterns: rearfoot and forefoot	5 years	NR	NA	Repetitive stress injuries
Burke et al. (2021)[20]	2564	19−42	M/F	20.3−24.6	Cross-country, recreational and competitive	Foot strike patterns: rearfoot, midfoot and forefoot	4 months−7 years	Yes	NR	Overall RRIs
Willwacher et al. (2022)[21]	16,370	NR	M/F	NR	NR	–	NR	NR	NA	Achilles tendinopathy, medial tibial stress syndrome, tibial stress fractures, plantar fasciitis, iliotibial band syndrome, PFP, patellar tendinopathy, hamstring tendinopathy
Ross et al. (2023)[22]	7353	37.3−46.5	M/F	NR	Novice, recreational and experienced	–	NR	Yes	Hip/Groin	Hip/groin RRIs
Louw and Dreary (2014)[35]	530	NR (age range: 18−50)	M/F	NR	Only runners	–	NR	Yes	NA	Iliotibial band syndrome

(*continued on next page*)

Table 3 (Continued)

Review	Participants characteristics						Follow-up period (min–max)	Injury		
	Total participants	Average age (min–max)	Gender	Mean BMI (min–max)	Running experience	Other characteristics		Definition	Anatomical area	Main diagnosis
Mann et al. (2016)[36]	669	NR	M/F	NR	Novice, recreational and experienced	–	NR	NR	NA	PFP, lower leg injuries, Achilles tendinopathy, iliotibial band syndrome, second metatarsal stress fracture, plantar fasciitis and RRIs

Abbreviations: BMI = body mass index; F = female; M = male; NA = not applicable; NR = not reported; PFP = patellofemoral pain; RRIs = running-related injuries.

Methodological differences between primary studies were found, which contributes to the heterogeneity of SRs and meta-analyses. This lack of consistency results in insufficient information to confirm the robustness and generalizability of the data. Several considerations should guide the design of future observational studies, including clarity in defining injuries,[38] specifying follow-up length, describing the studied population and outcomes, and thoroughly reporting confounding factors.[39] High-quality observational studies that address these aspects are necessary for advancing knowledge in this field.

The heterogeneity of SRs poses a challenge for comprehending the risk factors for injuries in running. Differences in the population investigated, injury definition, length of follow-up, and sample size across studies are some of the sources of heterogeneity. Most SRs studied mixed population, such as novice, recreational, competitive, and/or cross-country runners with very little attention given to running experience separately. Some SRs focused on a specific injury, such as patellofemoral pain,[34] iliotibial syndrome[35] or repetitive stress injuries,[19] but most reported on overall RRIs.[4,11,12,14,17] Five SRs did not report the length of follow-up.[14,21,22,35,36] Of those that report, the length of follow-up varied considerably among the SRs, ranging from 1 day[4] to 7 years.[20] Additionally, the definition of RRIs is different across individual studies and, consequently, in each SR. To address these issues, we suggest that future SRs should consider narrowing the scope and focusing on a single question, such as a specific injury or running experience.

It seems extremely important to have a clear definition for injury in future studies,[38] given that different definitions can lead to different estimates of effect for injury (OR, HR, or RR, for example), and incidence estimates. Lack of clarity in the description and definition of the outcomes is also an issue. Previous injury, for example, corresponds with “an injury 12 months before a new injury” in some studies, while others assess it as the history of injury, regardless of time.

RRIs seem to share common risk factors, particularly those associated with running/training characteristics and health-related lifestyle factors. These factors likely play a significant role in various types of RRIs. Conversely, biomechanical and morphological factors might be more specific to certain types of injuries, such as iliotibial band syndrome, patellofemoral pain syndrome, medial tibial stress syndrome, and Achilles tendinopathy. However, this remains speculative as, most primary studies and SRs did not address a specific injury (i.e. Achilles tendinopathy, patellofemoral pain, iliotibial band syndrome, and stress fractures). Instead, they grouped these injuries under the term “overall RRIs”, which complicates the identification of specific risk factors associated with each distinct injury.

Despite the efforts of researchers in the field, the main risk factors, or the set of risk factors for RRIs, remain unclear. The existing meta-analyses focus only on biomechanical factors. However, despite the large volume of information, the current evidence remains insufficient to conclusively identify biomechanical risk factors. Subsequent research efforts could

Table 4
Quality appraisal of the systematic reviews included on this umbrella review.

Study	AMSTAR tool																Total
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
Anderson et al. (2020)[19]	No	Partial yes	No	Partial yes	Yes	No	No	No	Yes	No	Yes	No	No	No	No	Yes	Critically low-quality review
Burke et al. (2021)[20]	Yes	Yes	Yes	Partial yes	Yes	No	No	Yes	Yes	No	0	0	Yes	Yes	0	Yes	Low-quality review
Louw and Dreary (2014)[35]	Yes	No	Yes	Partial yes	No	No	No	Partial yes	Yes	No	0	0	Yes	No	0	Yes	Critically low-quality review
Neal et al. (2016)[34]	Yes	No	Yes	No	Yes	No	No	Yes	Yes	No	No	No	No	No	No	Yes	Critically low-quality review
Saragiotto et al. (2014)[14]	Yes	No	Yes	Partial yes	Yes	Yes	No	Partial yes	Yes	No	0	0	Yes	Yes	0	Yes	Critically low-quality review
Van der Worp et al. (2015)[12]	Yes	Partial yes	Yes	Partial yes	Yes	No	No	Yes	Yes	No	0	0	Yes	Yes	0	Yes	Low-quality review
Van Gent et al. (2007)[4]	Yes	No	Yes	No	No	No	No	Yes	Yes	No	0	0	Yes	No	0	No	Critically low-quality review
Van Poppel et al. (2021)[11]	Yes	Yes	Yes	Partial yes	Yes	No	No	Yes	Yes	No	0	0	Yes	No	0	Yes	Low-quality review
Vanatta et al. (2020)[17]	Yes	Partial yes	Yes	Partial yes	Yes	No	No	Yes	Yes	No	Yes	Yes	Yes	No	No	Yes	Critically low-quality review
Willwacher et al. (2022)[21]	Yes	Partial yes	Yes	No	Yes	Yes	No	Yes	Yes	No	0	0	No	No	0	Yes	Critically low-quality review
Mann et al. (2016)[36]	No	No	Yes	Partial yes	Yes	No	No	Partial yes	Yes	No	0	0	Yes	No	0	Yes	Critically low-quality review
Peterson et al. (2022)[13]	Yes	Partial yes	Yes	Partial yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes	Low-quality review
Ross et al (2023)[22]	Yes	Partial yes	Yes	Partial yes	Yes	Yes	No	Yes	Yes	No	0	0	Yes	No	0	Yes	Low-quality review

Notes: (1) Did the research questions and inclusion criteria for the review include the components of PICO? (2) Did the report of the review contain an explicit statement that the review methods were established prior to the conduct of the review and did the report justify any significant deviations from the protocol? (3) Did the review authors explain their selection of the study designs for inclusion in the review? (4) Did the review authors use a comprehensive literature search strategy? (5) Did the review authors perform study selection in duplicate? (6) Did the review authors perform data extraction in duplicate? (7) Did the review authors provide a list of excluded studies and justify the exclusions? (8) Did the review authors describe the included studies in adequate detail? (9) Did the review authors use a satisfactory technique for assessing the risk of bias (RoB) in individual studies that were included in the review? (10) Did the review authors report on the sources of funding for the studies included in the review? (11) If meta-analysis was performed, did the review authors use appropriate methods for statistical combination of results? (12) If meta-analysis was performed, did the review authors assess the potential impact of RoB in individual studies on the results of the meta-analysis or other evidence synthesis? (13) Did the review authors account for RoB in primary studies when interpreting/discussing the results of the review? (14) Did the review authors provide a satisfactory explanation for, and discussion of, any heterogeneity observed in the results of the review? (15) If they performed quantitative synthesis did the review authors carry out an adequate investigation of publication bias (small study bias) and discuss its likely impact on the results of the review? (16) Did the review authors report any potential sources of conflict of interest, including any funding they received for conducting the review?
Abbreviations: AMSTAR = A MeaSurement Tool to Assess systematic Reviews; PICO = Population, Intervention, Comparation, and Outcome.

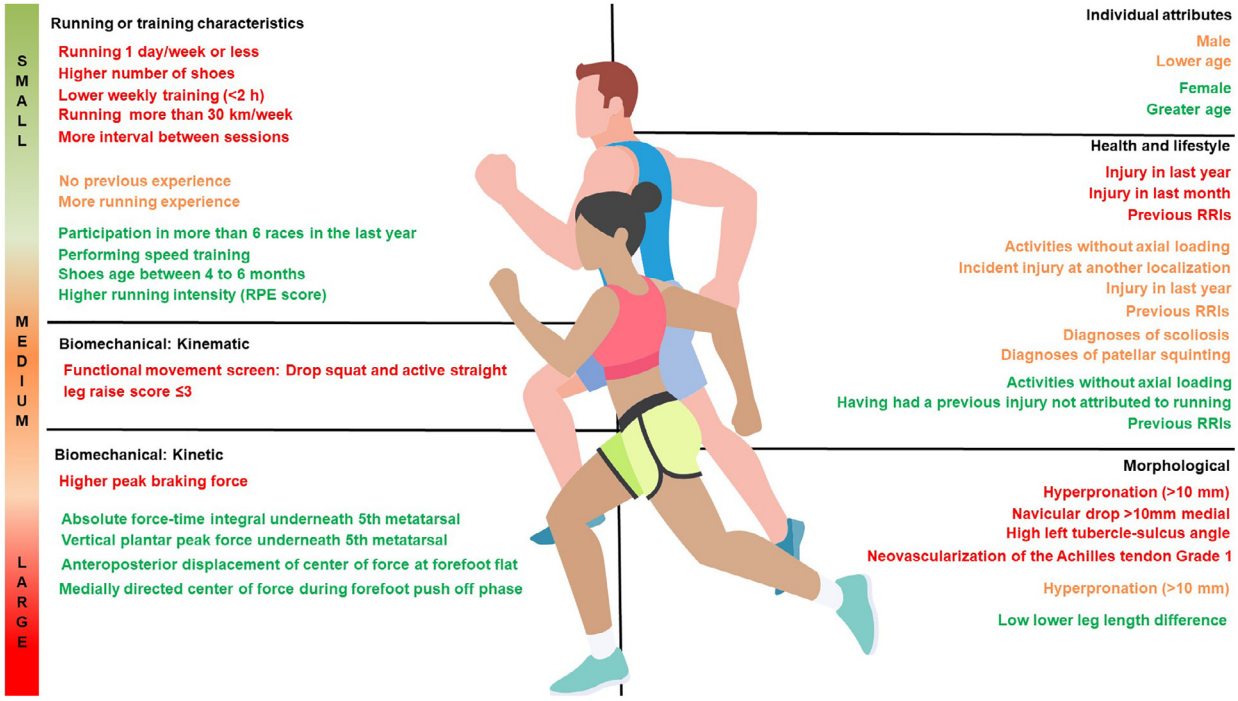


Fig 3. Overview of risk factors for RRIs considering only the systematic reviews with low quality, according to the classification of the effect size of the measures of risk (large, medium, and small effect sizes). Results are shown for studies that showed low quality ($n = 5$). RPE = rating of perceived exercise; RRIs = running-related injuries.

contribute to future meta-analyses incorporating individual attributes, such as running/training patterns, health and lifestyle indicators, and morphological risk factors.

The main strength of this work is to provide an overview of the current state of research on risk factors for RRIs. The comprehensiveness of the search and the qualitative approach used to report the current evidence must be acknowledged. We found a low overlap degree among the SRs, which reveals that few SRs concluded on the same available data. To the best of our knowledge, this was the first review to analyze the magnitude of effect sizes of the outcomes. This allows the best available evidence on injury prevention and aids readers in comprehending the associations magnitude and interpretating their practical significance. Interpreting the effect size for each available risk factor can significantly contribute to improved decision-making by professionals working with runners.

The comprehensive review allows the reader to explore diverse factors associated with running injuries. Interestingly, BMI[11,14] lacked a clear association, whereas heightened stature[4,14] exhibited a positive correlation with elevated injury risks. Examining lifestyle aspects unveiled alcohol consumption,[4] participation in other sports,[4] and certain medical histories,[4,22] including prior injuries, as contributory factors. Within the realm of running practices, increased race participation,[12] marathon running,[4] and initiation into running[4,11,14] demonstrated heightened injury risks. Weekly running distance[4,11,14] and volume[11] revealed medium to large associations, stressing the need for tailored exercise regimens. Biomechanical aspects, particularly focusing on hip[11,13,21,35] and pelvic[21] stability, as well as specific foot patterns,[11,13,17,20,21,36] emerged as significant in injury prevention. Finally, incorporating comprehensive functional movement screening,[11,13] particularly paying attention to a drop squat and the ability to lift one leg while lying and keeping the knee straight (active straight leg raise), can potentially aid in identifying individuals at a higher risk of overall RRIs. While these findings offer valuable insights, the need for further research is evident to establish conclusive relationships and inform targeted training strategies for injury risk reduction.

Of note, the quality assessment conducted in this umbrella review revealed an unsatisfactory scenario, with 8 out of the 13 SRs characterized by critically low-quality ratings. Fig. 3 excludes the findings from the 8 SRs with critically low quality. The supplementary materials present all the information extracted from the SRs, including that which was classified as critically low quality.

Some limitations must be considered in relation to our umbrella review. Our inclusion criteria resulted in a limited number of SRs included to compose this study. However, to summarize the available evidence and reach conclusive results regarding the risk factors for RRIs, a comprehensive and consistent research approach regarding RRIs definition, outcome variables, participants, and study design was required.

5. Conclusion

This umbrella SR provides a comprehensive overview about the risk factors for RRIs. Drawing from the outcomes of the low-quality SRs and associations with large effect sizes,

our findings indicate that running/training characteristics and health and lifestyle factors, as well as morphological and biomechanical aspects, are all implicated in elevating the risk of RRIs, emphasizing the multifactorial basis of injury incidence in running. The results of this umbrella review should be approached cautiously in clinical practice because none of the SRs included presented a rating of moderate or high confidence in quality.

Acknowledgments

This work was supported by the Programa de Bolsas Universitárias de Santa Catarina – Brasil (Uniedu) (Grant No. 14786 and Grant No. 16479) and the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES).

Authors' contributions

CKC conceived and designed the study, carried out the data collection, analysis and interpretation, and participated in writing (original draft, review, and editing); JMM carried out the data collection, analysis and interpretation, and participated in writing (original draft, review, and editing); FHD, MPdC, and HdBF provided critical feedback, and participated in writing (review and editing); CR conceived and designed the study, provided critical feedback, and participated in writing (review and editing). All authors have read and approved the final version of the manuscript, and agree with the order of presentation of the authors.

Competing interests

The authors declare that they have no competing interests.

Supplementary materials

Supplementary materials associated with this article can be found in the online version at doi:10.1016/j.jshs.2024.04.011.

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